

Day 2: TCP/IP Suite Theory (1 hour): What is the TCP/IP Suite? Layers of TCP/IP: Application, Transport, Internet, and Network Access. Practical (1 hour): Analyze packet headers to identify the TCP/IP layer using Wireshark. Task: Compare OSI and TCP/IP models with real-world examples.

Task 1 :- Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.

ANS :-

1. Application Layer (Top Level)

- What to look for in Wireshark: Hypertext Transfer Protocol (HTTP) or TLS.
- Header details: Shows the actual app request. For instance, GET / HTTP/1.1 asking Flipkart's web server to send over its homepage files.

2. Transport Layer

- What to look for in Wireshark: Transmission Control Protocol (TCP).
- Header details: Contains Source Port (a random port on your laptop, e.g., 51234) and Destination Port (443 for HTTPS or 80 for HTTP) to make sure data gets delivered to the right browser tab.

3. Internet Layer (Network Layer in OSI)

- What to look for in Wireshark: Internet Protocol Version 4 (IPv4).
- Header details: Contains your device's local Source IP (e.g., 192.168.1.5) and Flipkart's Destination IP (e.g., 163.53.78.89) to route the message across routers.

4. Network Access Layer (Data Link & Physical Layers)

- What to look for in Wireshark: Ethernet II (or IEEE 802.11 for Wi-Fi).
- Header details: Contains physical hardware addresses like Source MAC (your Wi-Fi card address) and Destination MAC (your home router address) to jump across local hardware cables/waves.

Task 2 :- Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).

ANS :-

OSI Layer	TCP/IP Layer	What It Does	App Example
Application	Application	Screen & buttons you see	Zomato: Viewing food menus
Presentation		Encrypts & translates data	WhatsApp: Securing messages
Session		Keeps connection open	Spotify: Staying logged in
Transport	Transport	Guarantees or speeds delivery	YouTube: Smooth video buffering
Network	Internet	Finds device IP addresses	Uber: Tracking driver location
Data Link	Network Access	Connects phone to router	Wi-Fi: Local wireless frames
Physical		Raw signals on hardware	5G: Radio waves to tower

Task 3 :- Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1-2 lines.

ANS :-

Source Port: Identifies the specific app sending the data on your device.

Destination Port: Identifies the receiving app on the target device (e.g., Port 443 for web traffic).

Sequence Number: Numbers each packet so the receiver can put them back in the correct order.

Acknowledgment (ACK) Number: Confirms to the sender which data bytes were received safely.

Header Length (Data Offset): Tells the receiver where the TCP header ends and actual app data begins.

Flags (Control Bits): Simple switches (like SYN, ACK, FIN) used to open, manage, or close a connection.

Window Size: Tells the sender how much data the receiver can handle at once (flow control).

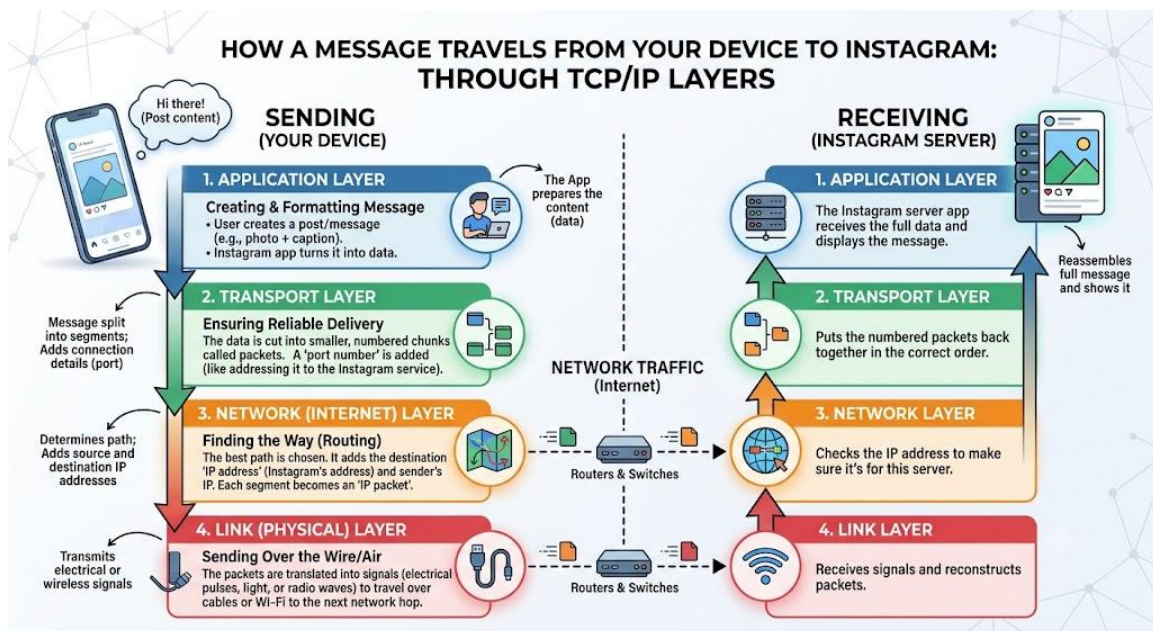
Checksum: Scans for errors to make sure data wasn't corrupted during transmission.

Urgent Pointer: Marks priority data that needs to be processed immediately.

Options & Padding: Extra optional settings used to tweak network connection rules.

Task 4 :- Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).

ANS :-



Task 5 :- Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app (like sending a WhatsApp message or booking a ticket on BookMyShow) it represents.

ANS :-

TCP/IP Layer: Application Layer (Top layer)

How to Identify It: In Wireshark, look at the Protocol column—it directly displays DNS (or TLSv1.3 / HTTP).

Real-World App Action: Opening the BookMyShow app and tapping on a movie—your phone uses DNS to look up the server address (api.bookmyshow.com) and HTTPS to fetch the movie list.